Dual-Space Calibrated Prototype Network for Open-Set Hyperspectral Image Classification

Shuai Ma, Jinrong He, *Member, IEEE*, and Zhaokui Li, *Member, IEEE*

***Abstract*—Open-set hyperspectral image (HSI) classification requires a model to recognize known land-cover categories while rejecting samples from unseen classes. This setting is difficult because HSIs have high spectral dimensionality, limited annotations, large intraclass variation, and strong interclass similarity. Many existing methods rely on closed-set confidence scores or combine heuristic anomaly cues, which makes the rejection boundary sensitive to dataset-specific score distributions. This article proposes a dual-space calibrated prototype network, called DSCP-Net, for open-set HSI classification. DSCP-Net learns two complementary feature spaces: one for local spectral–spatial structure and one for high-level semantic information. It uses dual-space multiprototype representations to describe the multimode structure of known classes, and employs a lightweight mixture-of-experts (MoEs) module for sample-adaptive feature transformation. For unknown detection, DSCP-Net adopts a distance-first inference rule in which calibrated prototype distances in the two spaces define the unknown score. A validation-based global calibration strategy determines the rejection threshold from known-class validation statistics. Experiments on four public hyperspectral datasets show that DSCP-Net improves the balance between known-class classification and unknown-class rejection over the compared closed-set and open-set methods. Further analysis shows that calibrated prototype-distance inference provides the clearest and most consistent contribution to the open-set gain, whereas the expert transformation and differentiable auxiliary objectives have smaller, dataset-dependent effects. The code is available at https://github.com/AIYAU/DSCP-Net**

***Index Terms*—Hyperspectral image (HSI) classification, mixture-of-experts (MoEs), open-set recognition (OSR), prototype learning, unknown class rejection, validation-based calibration.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is a fundamental task in remote sensing interpretation and has been widely used in precision agriculture, mineral exploration, environmental monitoring, target detection, and disaster assessment [1], [2], [3]. Compared with multispectral imagery, HSI records hundreds of contiguous narrow spectral bands, which makes it possible to distinguish land-cover materials with fine spectral differences [4], [5]. Early HSI classification methods usually relied on hand-crafted spectral or spatial features, such as dimensionality reduction, morphological profiles, and texture descriptors, followed by conventional classifiers including support vector machines and random forests [4], [5]. These methods established an important foundation for HSI interpretation, but their representation capacity is limited when handling high-dimensional data and complex nonlinear spectral–spatial distributions.

Deep learning has improved HSI classification by learning spectral–spatial representations in an end-to-end manner. Representative models include 3-D convolutional networks, residual networks, graph convolutional networks, and transformer-based architectures [3], [6], [7], [8], [9]. These methods have achieved strong performance under the closed-set assumption, where all test classes are assumed to be observed during training. In practice, however, remote sensing scenes rarely satisfy this assumption. A training set may cover the dominant crops, roofs, roads, or water bodies in one survey area, while a later image may contain newly built facilities, seasonal vegetation states, damaged materials, or rare land-cover types that were never annotated. In large-scale satellite monitoring, uncrewed aerial vehicle inspection, and disaster surveillance, forcing every such sample into a known class may produce apparently complete but unreliable maps, and the downstream cost can be higher than marking a region as uncertain [10], [11], [12], [13], [14].

This closed-set bias is especially problematic for HSI because classification maps are often used as intermediate products for later quantitative analysis. An unknown crop type labeled as a known crop, or a damaged surface labeled as a normal material, can propagate into estimates of area, yield, damage, or resource distribution. The error is then no longer a local pixel-level mistake. It changes the interpretation of the scene. A useful open-set classifier should therefore avoid two opposite failures: accepting unknown samples with high confidence and rejecting too many difficult known samples. This balance is harder than ordinary anomaly detection, because the model must preserve class discrimination among known categories while reserving the ability to say that a sample is outside the annotated label space.

Open-set recognition (OSR) addresses this issue by requiring a model to classify known samples and reject unknown ones. For HSI data, this task is particularly challenging. The

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same material may show different spectra under different illumination, moisture, atmospheric, or phenological conditions, whereas different materials may have similar reflectance patterns in certain spectral bands [1], [15], [16]. The high dimensionality of HSI data and the scarcity of labeled samples also make the learned feature space sparse and unstable, especially in few-shot settings [15], [17]. Mixed pixels, sensor noise, and cross-scene distribution shifts further blur the boundary between difficult known samples and true unknown samples. Open-set HSI classification is therefore not merely a matter of finding a rejection score. The score must remain meaningful when known classes are multimodal, unevenly sampled, and sometimes close to unknown classes. General OSR methods may produce unstable rejection boundaries when directly applied to this setting.

Existing OSR methods can be roughly grouped into confidence-based approaches, approaches based on extreme value theory, generative approaches, metric methods, and prototype-based approaches. Confidence-based methods reject samples using the maximum Softmax probability (MSP), but deep networks are often overconfident on unknown inputs [18], [19], [20], [21]. Methods based on extreme value theory, such as OpenMax, estimate open-set boundaries by modeling activation statistics, but their effectiveness depends on the stability of the learned feature distribution [11], [12]. Generative methods use reconstruction errors or synthesized unknown samples to support rejection, but generating physically plausible unknown HSI samples remains difficult due to the high dimensionality and spectral constraints of remote sensing data [22], [23], [24]. Metric methods and prototype-based methods provide more explicit geometric criteria by measuring the distance between a sample and known-class structures [25], [26], [27], [28], [29]. For HSI data, this geometric view is useful because an unknown sample should be far from the known spectral–spatial structures. The difficulty is that the known structures are rarely compact. A single compact prototype or a single feature space is often insufficient to represent the multimode intraclass distributions caused by illumination variation, spatial mixing, and material heterogeneity.

Another issue is threshold transfer across scenes and datasets. Even when an open-set score separates known and unknown samples on one dataset, its numerical scale may change with the sensor, spectral range, number of known classes, sampling ratio, or class composition. This sensitivity is severe in few-shot HSI classification, where the training set provides only a small view of the known-class distribution. A fixed threshold or an uncalibrated score fusion rule can therefore overfit a particular split. Validation-based calibration offers a more practical route: the rejection boundary is estimated from held-out known-class statistics, while unknown test samples remain unseen during threshold selection. Such a design matches the deployment condition more closely and reduces the risk of using unknown-class information implicitly.

Recent studies have explored open-set HSI classification from several directions. Few-shot calibration and unknown-aware training address the limited-label setting [14], [30]. Reconstruction-based and semantic-reconstruction strategies reject unknown samples through reconstruction or category-consistency errors [31], [32]. Contrastive and graph-based methods improve spectral–spatial separability [33], [34], while frequency-domain, global-distribution, and language-enhanced methods introduce HSI-specific priors [35], [36], [37]. Open-set domain adaptation is related but also considers source-target domain shifts [38], [39], [40], [41]. These studies have made important progress, but two practical issues remain. First, known-class classification and unknown rejection are often handled in the same representation space [28], [30], [33]. In HSI, local spectral–spatial patterns and high-level semantic discrimination often require different feature geometries. Second, several methods combine confidence, distance, entropy, reconstruction error, or other anomaly cues [12], [22], [31], [42]. Such fusion can be effective on a given split, but the scale and reliability of these cues vary across datasets, so the threshold may become tied to dataset-specific score distributions.

Motivated by these observations, this article proposes a dual-space calibrated prototype network, called DSCP-Net, for open-set HSI classification. DSCP-Net separates local structural assignment from semantic discrimination by learning a routing space and an image space. It then describes each known class with multiple prototypes in both spaces, so that open-set inference can be based on distances to structured known-class modes rather than on Softmax confidence alone. A sparse mixture-of-experts (MoEs) module provides sample-adaptive transformation for intraclass variation and local modes [43], [44]. Finally, DSCP-Net estimates validation statistics from known-class samples and applies quantile-based global calibration to set the rejection boundary.

The main contributions of this article are summarized as follows.

1) We propose DSCP-Net for open-set HSI classification. It integrates dual-space representation learning, multiprototype known-class modeling, conditional expert transformation, and validation-based calibration into a unified distance-first inference framework. Instead of treating unknown detection as confidence thresholding applied after training, DSCP-Net formulates OSR as a unified process of known-class structure learning, distance-first unknown scoring, and calibrated rejection.

2) We design a dual-space multiprototype representation mechanism for modeling known-class structures. The routing space is used to describe local spectral–spatial structural cues and guide expert assignment, whereas the image space is used for semantic discrimination and prototype-based rejection. Multiprototype modeling explicitly represents intraclass local modes, while the conditional expert transformation provides an auxiliary sample-adaptive representation path whose marginal HOS effect is evaluated separately and varies across scenes.

3) We introduce a calibrated distance-first open-set inference strategy. The unknown score is computed from dual-space prototype distances and normalized using only known-class validation statistics. A quantile-based threshold is then used to determine the rejection boundary, providing an interpretable mechanism to control the tradeoff between known-class preservation and unknown-class rejection across different hyperspectral scenes.

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II. Related Work

A. HSI Classification

HSI classification assigns land-cover labels to pixels by using both spectral signatures and spatial context. Traditional HSI classifiers usually combine hand-crafted spectral or spatial features with shallow models such as support vector machines and random forests [4], [5]. Convolutional neural networks and 3-D convolutional networks have become widely used in HSI classification because they can jointly model spectral and spatial information [1], [2], [6]. Generative adversarial networks, graph convolutional networks, graph reasoning methods, and graph-based variants have also been studied to improve feature representation under limited supervision [7], [8], [9], [45]. Transformer-based models further extend this line of work by capturing long-range spectral–spatial dependencies and global contextual information [3], [34]. Recent attention- and segmentation-oriented methods, such as central attention, multiarea target attention, and SegHSI, further show the importance of structured spatial–spectral context modeling for HSI interpretation [46], [47], [48].

These methods have improved closed-set classification accuracy, but they usually assume that the training and testing stages share the same label space. This assumption is restrictive in remote sensing applications, where previously unseen land-cover types may appear at test time. When a closed-set classifier encounters an unknown sample, it still produces a known-class label, often with high confidence. Open-set HSI classification therefore requires discriminative known-class representations and reliable mechanisms for rejecting unknown samples.

B. OSR and Out-of-Distribution (OOD) Detection

OSR controls the empirical classification risk on known classes and the open-space risk caused by unknown classes [10], [13]. A common baseline rejects samples according to the MSP, but neural networks may still assign high confidence to OOD or unknown samples [18], [21]. OpenMax extends the closed-set classifier by fitting activation statistics with extreme value theory and reserving probability mass for unknown classes [12]. Other methods use reconstruction errors, synthetic unknown samples, virtual-logit or OOD scores, and adversarial learning to define more reliable open-set decision boundaries [22], [23], [24], [42]. Metric-based and prototype-based methods measure how far a test sample deviates from known-class representations, providing a more interpretable geometric criterion for rejection [26], [27], [28].

OOD detection is closely related to OSR, but the two settings differ in their objectives. OOD detection mainly determines whether a sample comes from the training distribution, whereas OSR must also assign accepted known samples to the correct known categories. For open-set HSI classification, increasing the rejection rate alone is insufficient, because difficult known samples may also be rejected. An effective method should therefore balance known-class classification and unknown-class rejection instead of optimizing either side independently.

C. Open-Set HSI Classification

Open-set HSI classification has received increasing attention in recent years. Early efforts investigated few-shot OSR with calibration networks to address the limited-label setting [30]. Reconstruction modeling and semantic reconstruction were then introduced to define unknown rejection cues from reconstruction quality or category consistency [14], [31], [32], [49]. Supervised contrastive learning, reciprocal-point learning, graph contrastive learning, and orientational prototype modeling further improve feature geometry for separating known and unknown classes [33], [34], [50], [51]. Recent task-adaptive transformers, frequency-domain methods, global distribution modeling, uncertainty-aware strategies, and language-enhanced contrastive learning also show the value of HSI-specific structure modeling [35], [36], [37], [52], [53]. Open-set domain adaptation is related to this topic but focuses on domain shifts between training and testing scenes, using reconstruction discrepancy, feature alignment, adversarial learning, manifold mapping, or multimodal progressive learning [38], [39], [40], [41], [54]. Universal domain adaptation further relaxes fixed source–target label-set assumptions; recent remote sensing methods identify shared and unknown categories through confidence–consistency–certainty criteria, cluster-based semantic pairing, or dual-classifier consistency with cross-domain feature mixup [55], [56], [57].

Despite this progress, two issues remain insufficiently studied. Many methods perform known-class classification and unknown rejection in the same representation space. For HSI data, local spectral–spatial variations and high-level semantic discrimination may require different representation emphases, and using a single space can lead to unstable rejection boundaries. Some methods also rely on multiple anomaly scores, but the relative reliability of these scores can vary considerably across datasets. Without proper calibration, score fusion may introduce additional uncertainty that is specific to each dataset. In contrast, this article first builds structured known-class representations and then uses calibrated prototype distances as the main criterion for open-set inference.

D. Prototype Learning and Conditional Expert Transformation

Prototype learning represents each class with one or more vectors in the feature space. This formulation is useful for few-shot and open-set scenarios because distances to prototypes provide an interpretable measure of class membership [27], [28], [29], [49], [58]. However, a single prototype implicitly assumes that each class forms a compact unimodal distribution, which often does not hold for HSI data. Due to illumination variation, spatial mixing, and phenological changes, the same land-cover category may contain multiple stable intraclass modes. Multiprototype modeling is therefore better suited to describing complex known-class distributions and measuring how far a test sample deviates from known-class structures.

MoEs models use a routing mechanism to send different samples to different expert transformations [43], [44], [59]. This conditional computation can improve representation

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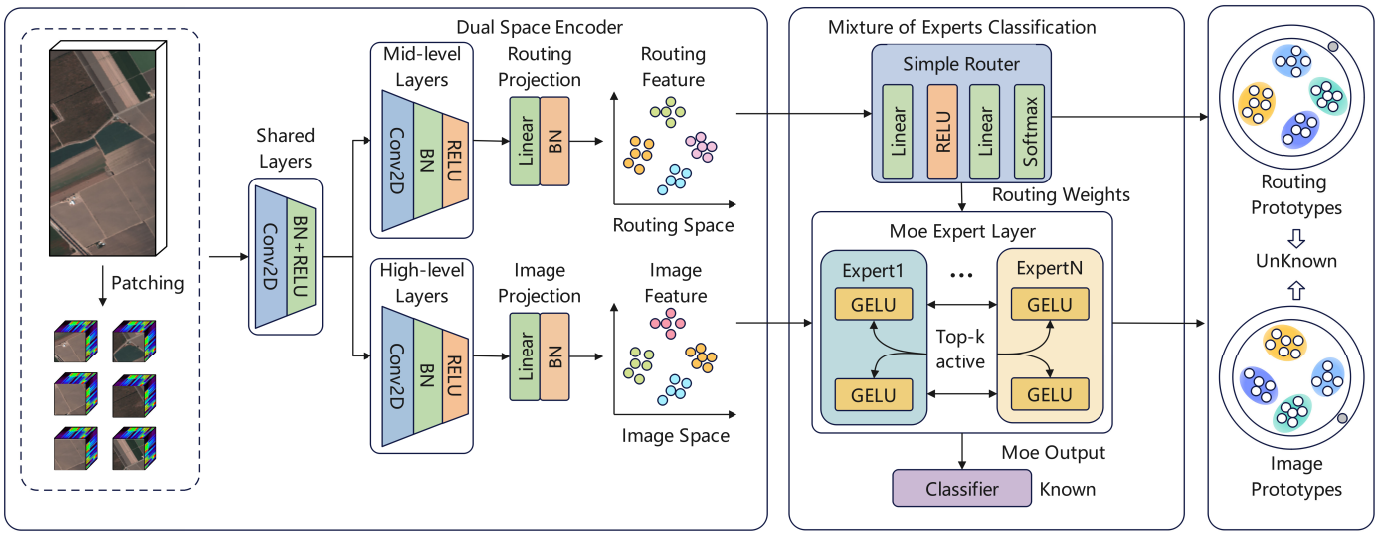


Fig. 1. Overall architecture of DSCP-Net. Given an input hyperspectral patch, the shared encoder extracts two complementary representations, i.e., the routing-space feature and the image-space feature. The routing-space feature guides Top-K expert selection, while the image-space feature is transformed by the activated experts for known-class classification. During open-set inference, dual-space prototype distances are calibrated by known-class validation statistics to reject unknown samples.

flexibility and encourage expert specialization. In DSCP-Net, MoE is used as a conditional transformation module rather than as a direct unknown detector. The routing space generates expert assignment probabilities, while the image space is transformed by the activated experts for semantic discrimination and prototype-based rejection. This design provides an additional sample-conditional representation path before calibrated prototype-distance inference.

III. Method

This section presents DSCP-Net. The key idea is to model known-class structures in two complementary feature spaces and reject unknown samples by calibrated prototype-distance inference.

As shown in Fig. 1, DSCP-Net consists of three main components. A dual-space encoder extracts routing-space and image-space representations from each hyperspectral patch. Dual-space multiprototype representations describe known-class structures in both spaces. A sparse MoEs module performs sample-conditional feature transformation as an auxiliary representation component, and validation-based calibration derives the distance-first open-set decision rule.

A. Problem Formulation

Let the training set be $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$ with $y_i \in \{1, 2, \ldots, C\}$, where $x_i \in \mathbb{R}^{B\times P\times P}$ is a hyperspectral patch centered at the ith labeled pixel, B denotes the number of spectral bands, $P \times P$ is the spatial neighborhood size, and C is the number of known classes in training. Only known-class samples are provided for supervised training. At test time, samples from unknown classes may appear, so the test label space is $y_i^{\text{test}} \in \{1, 2, \ldots, C, C+1\}$, where $C+1$ denotes the unknown class. The goal is to classify known-class samples correctly and reject unknown-class samples. This protocol considers OSR within each dataset split. It does not assume a separate labeled source domain and unlabeled target domain, and therefore it is different from open-set domain adaptation.

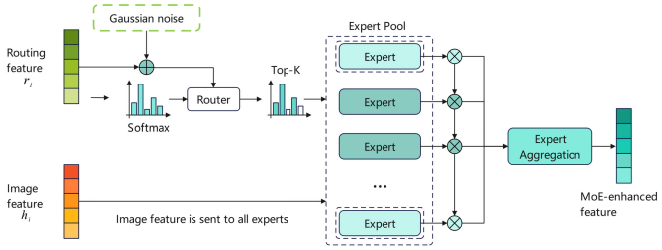


Fig. 2. Sparse MoE routing module. The routing-space feature $\mathbf{r}_i$ produces noisy routing logits and sparse Top-K expert weights, while the image-space feature $\mathbf{h}_i$ is transformed by the activated experts. The final representation $\mathbf{z}_i$ is computed as the weighted aggregation of the selected expert outputs.

The distributional assumption applies only to the within-scene protocol. Known-class pixels used for training, validation, and testing come from the same HSI and thus share the sensor, acquisition conditions, and preprocessing. We treat their class-conditional distributions as approximately matched, without assuming that spatially neighboring pixels are strictly independent. Formally, the protocol assumes $p_{\text{train}}(x \mid y = c) \approx p_{\text{test}}(x \mid y = c)$ for each known class $c \in \{1, \ldots, C\}$. The test label support additionally contains the unknown category $C+1$, so the intended train–test difference is label-space expansion rather than covariate or domain shift among the known classes. If the sensor, location, season, or acquisition conditions change, this assumption may fail and the validation-calibrated threshold may need to be recalibrated. Cross-domain OSR is outside the scope of this study.

B. Overall Framework

Given an input sample x_i, the shared encoder extracts two complementary features: the routing-space feature r_i and the image-space feature h_i. The routing-space feature r_i generates

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expert routing weights, while the image-space feature h_i is transformed by the activated experts into the representation z_i. The resulting representation then produces the closed-set classification logits ℓ_i. The forward process is written compactly as

$$\begin{aligned} x_i &\xrightarrow{\text{Encoder}} (r_i, h_i) \\ r_i &\xrightarrow{\text{Router}} \hat{p}_i \\ (h_i, \hat{p}_i) &\xrightarrow{\text{MoE}} z_i \xrightarrow{\text{Classifier}} \ell_i. \end{aligned} \tag{1}$$

During open-set inference, DSCP-Net does not use Softmax confidence directly for unknown detection. It instead computes an unknown score S_i from the dual-space multiprototype structure and applies global calibration using validation statistics to obtain the final open-set prediction $\hat{y}_i$.

C. Dual-Space Representation Learning

A hyperspectral sample carries both local spectral–spatial structure and high-level category semantics. When classification and rejection rely on a single feature space, local structural variations may be mixed with semantic discrimination, which can make open-set boundaries unstable. DSCP-Net therefore constructs two complementary spaces. The routing space captures mid-level local structural cues for expert routing, whereas the image space captures high-level category semantics for expert transformation and classification. In the implementation, the shared encoder first produces a common feature map; the routing feature is obtained from a mid-level branch with global average pooling, while the image-space feature is produced by a high-level branch with adaptive average pooling. Here, "image space" denotes a latent semantic feature space derived from the hyperspectral patch, not the original pixel-domain or spatial-domain image plane.

Let $f_{\text{shared}}(\cdot)$ be the shared feature extractor, and let $f_{\text{mid}}(\cdot)$ and $f_{\text{high}}(\cdot)$ be the mid-level and high-level feature extraction modules, respectively. The two features are computed as

$$\begin{aligned} u_i &= f_{\text{shared}}(x_i) \\ m_i &= f_{\text{mid}}(u_i) \\ r_i &= g_r(\text{GAP}(m_i)) \\ h_i &= g_h\left(f_{\text{high}}(m_i)\right) \end{aligned} \tag{2}$$

where $g_r(\cdot)$ and $g_h(\cdot)$ are projection heads, GAP$(\cdot)$ denotes global average pooling, and $r_i, h_i \in \mathbb{R}^d$. In implementation, $f_{\text{high}}(\cdot)$ contains the high-level convolutional block followed by adaptive average pooling, and both projected features are batch-normalized before routing, prototype updating, and open-set scoring. This dual-space design describes known-class distributions from complementary local-structural and semantic perspectives.

D. MoE-Enhanced Known-Class Discriminative Modeling

Known hyperspectral classes often contain large intraclass variation and multiple local modes. A single transformation path may average these modes and weaken fine-grained class boundaries. DSCP-Net uses an MoEs mechanism to learn sample-conditional transformations for known-class representation. MoE is not used as a direct unknown detector; it supplies an auxiliary representation path before distance-based open-set inference. The sparse MoE routing process is illustrated in Fig. 2, where the routing-space feature produces Top-K expert weights and the image-space feature is transformed by the selected experts.

The routing-space feature r_i generates the expert routing distribution. Given M experts, the routing logits, noisy logits, and soft routing distribution are computed as

$$\begin{aligned} a_i &= W_{\text{router}} r_i + b_{\text{router}} \\ \tilde{a}_i &= a_i + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}\left(0, \sigma^2 I\right) \\ p_i &= \text{softmax}\left(\frac{\tilde{a}_i}{\tau}\right), \qquad p_i \in \mathbb{R}^M \end{aligned} \tag{3}$$

where p_{im} is the routing weight from sample i to expert m and τ is the temperature parameter.

To reduce computation and encourage expert specialization, DSCP-Net adopts a Top-K sparse activation strategy. Let $\mathcal{S}_i$ denote the set of activated experts for sample i, with $|\mathcal{S}_i| = K$. The selected expert weights are renormalized and used to aggregate the expert outputs as

$$\begin{aligned} \hat{p}_{im} &= \frac{p_{im}}{\sum_{j \in \mathcal{S}_i} p_{ij}}, \quad m \in \mathcal{S}_i \\ z_i &= \sum_{m \in \mathcal{S}_i} \hat{p}_{im} E_m(h_i). \end{aligned} \tag{4}$$

When MoE is disabled, this module reduces to $z_i = h_i$.

E. Dual-Space Multiprototype Representation

A single prototype implicitly assumes that each class forms a compact unimodal distribution. This assumption often does not hold for hyperspectral data, where the same land-cover category may show multiple stable spectral–spatial patterns because of illumination changes, material variability, and spatial context. DSCP-Net therefore maintains multiple prototypes for each known class in both the routing space and the image space

$$\begin{aligned} \mathcal{P}_c^r &= \left\{p_{c,1}^r, p_{c,2}^r, \ldots, p_{c,K_p}^r\right\} \\ \mathcal{P}_c^h &= \left\{p_{c,1}^h, p_{c,2}^h, \ldots, p_{c,K_p}^h\right\} \end{aligned} \tag{5}$$

where K_p is the number of prototypes per class and $p_{c,k}^r, p_{c,k}^h \in \mathbb{R}^d$.

The prototypes are implemented as running buffers and are updated using only known-class training samples. For each mini-batch, samples of class c are first grouped into K_p local modes. When enough samples are available, K-means is applied in the routing space and the same cluster assignment is used to update the corresponding routing-space and image-space prototypes. When the mini-batch contains too few samples for reliable clustering, samples are assigned to the nearest existing prototype, or to the first prototype before the class has been initialized. For each nonempty group $\mathcal{B}_{c,k}$, the prototypes are updated by exponential moving average

$$p_{c,k}^s \leftarrow \mu p_{c,k}^s + (1-\mu)\frac{1}{|\mathcal{B}_{c,k}|}\sum_{i \in \mathcal{B}_{c,k}} v_i^s, \quad s \in \{r, h\} \tag{6}$$

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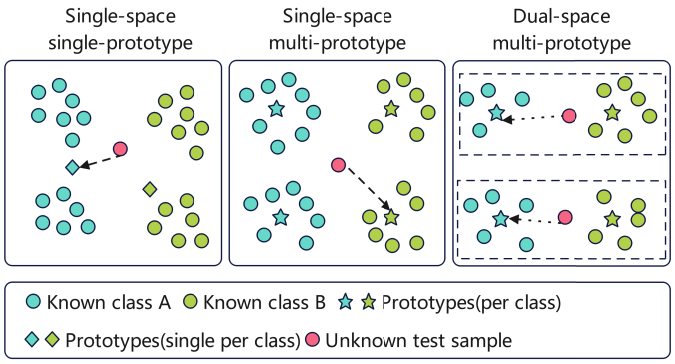


Fig. 3. Geometric interpretation of dual-space multiprototype modeling. A single prototype may over-compress multimode known-class distributions, whereas multiple prototypes describe different intraclass local modes. The dual-space design evaluates each test sample in both the routing space and the image space, providing a structured basis for prototype-distance-based open-set rejection.

where $v_i^r = r_i$, $v_i^h = h_i$, and μ is the momentum coefficient. This update makes K_p an explicit prototype granularity parameter rather than an additional classifier output. The classification loss backpropagates through the complete trainable network. The prototype contrastive and dual-space alignment losses update the feature paths on which they are defined, and the soft-routing importance balance loss updates the routing branch. The prototype buffers receive no gradients and instead follow the EMA rule in (6).

For sample i, the routing-space and image-space distances to class c are

$$d_r(i,c) = \min_{k\in\{1,\ldots,K_p\}} \left(1-\cos\left(r_i, p_{c,k}^r\right)\right)$$
$$d_h(i,c) = \min_{k\in\{1,\ldots,K_p\}} \left(1-\cos\left(h_i, p_{c,k}^h\right)\right). \quad (7)$$

The nearest classes in the two spaces are

$$\hat{y}_i^r = \arg\min_{c\in\{1,\ldots,C\}} d_r(i,c) \quad \hat{y}_i^h = \arg\min_{c\in\{1,\ldots,C\}} d_h(i,c). \quad (8)$$

As shown in Fig. 3, the multiprototype mechanism represents each known class with several local modes rather than a single center. An unknown sample should therefore deviate from all learned known-class modes in both the routing space and the image space.

F. Calibrated Distance-First Open-Set Inference

A straightforward open-set baseline rejects samples according to the MSP. However, this confidence-based criterion is often unreliable for HSIs because spectrally similar unknown classes may still receive high closed-set confidence. DSCP-Net therefore uses a distance-first inference strategy, in which the open-set score is determined by the geometric deviation from learned known-class prototypes rather than by classifier confidence alone.

For each test sample, DSCP-Net searches for the nearest known-class structure independently in the routing space and the image space. The routing-space distance measures deviation from local spectral–spatial structures, whereas the image-space distance measures deviation from high-level semantic structures. The two distances are averaged to form the raw unknown score

$$d_i^{\text{avg}} = \frac{d_r\left(i,\hat{y}_i^r\right) + d_h\left(i,\hat{y}_i^h\right)}{2}. \quad (9)$$

This score measures how far the sample is from known-class patterns in both the local structural space and the high-level semantic space. A sample far from the nearest known prototypes in both spaces is more likely to be unknown.

Prototype distances may have different numerical scales across datasets, so a fixed threshold on the raw distance score can lead to unstable open-set boundaries. DSCP-Net therefore uses a validation-based global calibration strategy. Only known-class validation samples are used to estimate the mean and standard deviation of the distance scores, and each test score is normalized by z-score standardization

$$S_i = \frac{d_i^{\text{avg}} - \mu_d}{\sigma_d + \varepsilon} \quad (10)$$

where μ_d and σ_d are the mean and standard deviation of the calibration-set distance distribution, respectively, and ε is a small constant used to avoid division by zero. The calibration set $\mathcal{D}_{\text{val}} \subset \mathcal{D}_{\text{train}}$ contains only known-class samples. The standardized unknown scores of the calibration samples are $\{S_i^{\text{val}}\}_{i=1}^{N_{\text{val}}}$. The global rejection threshold is estimated with the quantile function

$$\theta = Q_q\left(\{S_i^{\text{val}}\}_{i=1}^{N_{\text{val}}}\right) \quad (11)$$

where $Q_q(\cdot)$ denotes the qth quantile. We set q to 0.95 by default. This strategy determines the rejection boundary from the known-class validation distribution and avoids manually fixed thresholds across datasets.

Open-Set Decision Rule: For a test sample x_i, the model obtains the closed-set prediction as $\hat{y}_i^{\text{cls}} = \arg\max_{c\in\{1,\ldots,C\}} \ell_{i,c}$. The standardized unknown score is then compared with the calibrated threshold

$$\hat{y}_i = \begin{cases} C+1, & S_i > \theta \\ \hat{y}_i^{\text{cls}}, & S_i \le \theta. \end{cases} \quad (12)$$

Thus, even when the classifier assigns a sample to a known class, the sample can still be rejected if it deviates strongly from the dual-space prototype structure.

G. Training Objective

DSCP-Net is trained with known-class labels. The gradient-based objective used in all reported experiments is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{proto}}\mathcal{L}_{\text{proto}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}} + \lambda_{\text{bal}}\mathcal{L}_{\text{bal}}. \quad (13)$$

Here, $\mathcal{L}_{\text{cls}}$ is the classification term, and $\mathcal{L}_{\text{proto}}$, $\mathcal{L}_{\text{align}}$, and $\mathcal{L}_{\text{bal}}$ are differentiable auxiliary terms. The EMA prototype update is not part of this objective, and validation-based calibration is applied only during inference. We do not expect every auxiliary term to improve HOS on every scene.

1) Classification Loss: Known-class classification is trained with the standard cross-entropy loss

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp\left(\ell_{i,y_i}\right)}{\sum_{c=1}^{C}\exp\left(\ell_{i,c}\right)}. \quad (14)$$

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Here, $\ell_{i,c}$ denotes the classifier logit of sample i for known class c.

2) Prototype Contrastive Loss: The prototype contrastive loss shapes the dual-space prototype structure. For a feature v_i^s in space $s \in \{r, h\}$, where $v_i^r = r_i$ and $v_i^h = h_i$, all prototypes are first normalized and the similarity logits are computed as

$$\alpha_{i,c,k}^s = \frac{\cos\left(v_i^s, p_{c,k}^s\right)}{\tau_c}. \tag{15}$$

The contrastive target assigns equal probability to the K_p prototypes of the ground-truth class and zero probability to prototypes of other classes. The loss in space s is

$$\mathcal{L}_{\text{proto}}^s = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K_p}\frac{1}{K_p}\log\frac{\exp\left(\alpha_{i,y_i,k}^s\right)}{\sum_{c=1}^{C}\sum_{j=1}^{K_p}\exp\left(\alpha_{i,c,j}^s\right)}. \tag{16}$$

The final prototype contrastive loss is

$$\mathcal{L}_{\text{proto}} = \frac{1}{2}\left(\mathcal{L}_{\text{proto}}^r + \mathcal{L}_{\text{proto}}^h\right). \tag{17}$$

This term encourages each sample to be close to the prototype set of its ground-truth class in both spaces and imposes explicit structural constraints during training.

3) Dual-Space Alignment Loss: The alignment loss improves semantic consistency between the two feature spaces at the class-distribution level rather than directly forcing r_i and h_i to become identical. Let $\bar{p}_c^r$ and $\bar{p}_c^h$ denote the mean prototypes of class c in the two spaces. The class-probability distributions induced by the two spaces are

$$q_i^r = \text{softmax}\left(\frac{\left[\cos\left(r_i, \bar{p}_1^r\right), \ldots, \cos\left(r_i, \bar{p}_C^r\right)\right]}{\tau_a}\right)$$
$$q_i^h = \text{softmax}\left(\frac{\left[\cos\left(h_i, \bar{p}_1^h\right), \ldots, \cos\left(h_i, \bar{p}_C^h\right)\right]}{\tau_a}\right). \tag{18}$$

The alignment loss is the symmetric KL divergence between these two distributions:

$$\mathcal{L}_{\text{align}} = \frac{1}{2N}\sum_{i=1}^{N}\left[D_{\text{KL}}\left(q_i^h \| q_i^r\right) + D_{\text{KL}}\left(q_i^r \| q_i^h\right)\right]. \tag{19}$$

This term does not force the two spaces to become identical. It instead improves their semantic consistency while preserving their complementary roles.

4) Soft-Routing Importance Balance Loss: To discourage a small number of experts from dominating the soft routing distribution, DSCP-Net uses differentiable importance balancing. The importance of expert m is

$$I_m = \frac{1}{N}\sum_{i=1}^{N} p_{im}. \tag{20}$$

The soft-routing balance loss is

$$\mathcal{L}_{\text{bal}} = \sum_{m=1}^{M}\left(I_m - \frac{1}{M}\right)^2. \tag{21}$$

This objective regularizes the average routing probabilities while remaining differentiable with respect to the router parameters.

TABLE I

DATASET SPLIT OF THE HOUSTON DATASET UNDER THE OPEN-SET PROTOCOL. KNOWN: CLASSES 1–12; UNKNOWN: CLASSES 13–15

Class	Class Name	Samples	Training	Testing
1	Healthy Grass	1,251	20	1,231
2	Stressed Grass	1,254	20	1,234
3	Synthetic Grass	697	20	677
4	Tree	1,244	20	1,224
5	Soil	1,242	20	1,222
6	Water	325	20	305
7	Residential	1,268	20	1,248
8	Commercial	1,244	20	1,224
9	Road	1,252	20	1,232
10	Highway	1,227	20	1,207
11	Railway	1,235	20	1,215
12	Parking Lot 1	1,233	20	1,213
13–15	Unknown	1,557	–	1,557
Total		15,029	240	14,789

TABLE II

DATASET SPLIT OF THE IP DATASET UNDER THE OPEN-SET PROTOCOL. KNOWN: CLASSES 1–8; UNKNOWN: CLASSES 9–16

Class	Class Name	Samples	Training	Testing
1	Alfalfa	46	20	26
2	Corn-notill	1428	20	1408
3	Corn-mintill	830	20	810
4	Corn	237	20	217
5	Grass-Pasture	483	20	463
6	Grass-Trees	730	20	710
7	Grass-Pasture-mowed	28	20	8
8	Hay-windrowed	478	20	458
9–16	Unknown	5989	–	5989
Total		10249	160	10089

TABLE III

DATASET SPLIT OF THE LK DATASET UNDER THE OPEN-SET PROTOCOL. KNOWN: CLASSES 1–8; UNKNOWN: CLASS 9

Class	Class Name	Samples	Training	Testing
1	Corn	34511	20	34491
2	Cotton	8374	20	8354
3	Sesame	3031	20	3011
4	Broad-leaf Soybean	63212	20	63192
5	Narrow-leaf Soybean	4151	20	4131
6	Rice	11854	20	11834
7	Water	67056	20	67036
8	Roads and Houses	7124	20	7104
9	Unknown	5229	–	5229
Total		204542	160	204382

IV. EXPERIMENTS

A. Datasets and Open-Set Protocol

We evaluate DSCP-Net under few-shot open-set conditions on four public hyperspectral datasets: Houston, Indian Pines (IP), WHU-Hi-LongKou (LK), and Pavia University (UP). Tables I–IV summarize the class-wise splits used in the open-set protocol.

1) Houston Dataset: The Houston dataset was acquired by the ITRES CASI-1500 sensor over the University of Houston campus and the surrounding urban area. It contains 144 spectral bands with a spatial resolution of 2.5 m. Under the

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TABLE IV
DATASET SPLIT OF THE UP DATASET UNDER THE OPEN-SET PROTOCOL. KNOWN: CLASSES 1–8; UNKNOWN: CLASS 9

Class	Class Name	Samples	Training	Testing
1	Asphalt	6631	20	6611
2	Meadows	18649	20	18629
3	Gravel	2099	20	2079
4	Trees	3064	20	3044
5	Painted metal sheets	1345	20	1325
6	Bare Soil	5029	20	5009
7	Bitumen	1330	20	1310
8	Self-Blocking Bricks	3682	20	3662
9	Unknown	947	–	947
Total		42776	160	42616

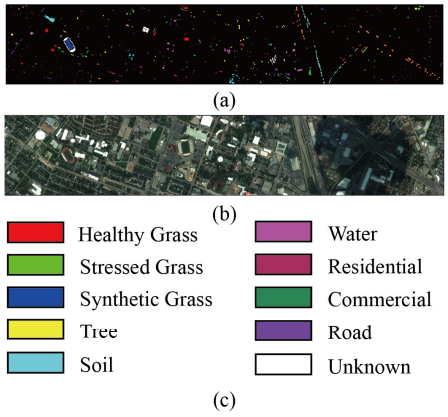


Fig. 4. (a) Ground-truth map, (b) false-color image, and (c) class legend of the Houston dataset.

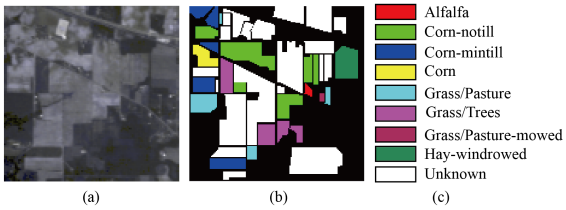


Fig. 5. (a) False-color image, (b) ground-truth map, and (c) class legend of the IP dataset.

open-set protocol, classes 1–12 are used as known classes, and classes 13–15 are treated as unknown classes. The false-color image, ground-truth map, and class legend are shown in Fig. 4.

2) IP Dataset: The IP dataset was collected by the AVIRIS sensor over an agricultural area in northwestern Indiana, USA. After removing water-absorption and low-SNR bands, 200 spectral bands are retained. The dataset contains many spectrally similar crop categories, making it challenging for OSR. Classes 1–8 are used as known classes, and classes 9–16 are treated as unknown classes. The false-color image, ground-truth map, and class legend are shown in Fig. 5.

3) LK Dataset: The LK dataset was acquired by an uncrewed aerial vehicle over an agricultural area in Hubei Province, China. It contains 270 spectral bands and has a spatial resolution of 0.463 m. Classes 1–8 are used as known classes, and class 9 is treated as the unknown class. The false-color image, ground-truth map, and class legend are shown in Fig. 6.

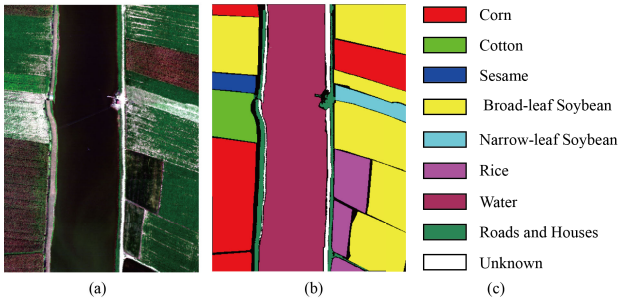


Fig. 6. (a) False-color image, (b) ground-truth map, and (c) class legend of the LK dataset.

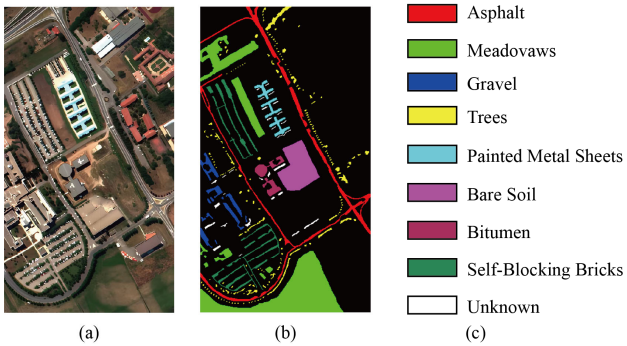


Fig. 7. (a) False-color image, (b) ground-truth map, and (c) class legend of the UP dataset.

4) UP Dataset: The UP dataset was acquired by the ROSIS sensor over the University of Pavia, Italy. After removing noisy bands, 103 spectral bands are retained, and the image size is 610 × 340 with a spatial resolution of approximately 1.3 m. The scene mainly includes urban land-cover categories such as asphalt, meadows, gravel, trees, painted metal sheets, bare soil, bitumen, self-blocking bricks, and shadows. Classes 1–8 are used as known classes, and class 9 is treated as the unknown class. The false-color image, ground-truth map, and class legend are shown in Fig. 7.

B. Experimental Settings

All experiments are implemented in PyTorch. DSCP-Net experiments are conducted on NVIDIA V100 GPUs, and the reproduced baseline experiments are run on NVIDIA V100 GPUs. Each HSI is normalized band by band before training. For each labeled pixel, we extract a 9×9 spatial neighborhood centered on the pixel as the input patch. For each known class, 20 labeled samples are randomly selected as the labeled training-calibration pool. This pool is internally split into a training subset and a known-class validation subset for threshold calibration, and the remaining known-class samples, together with all unknown-class samples, are used only for testing. The main comparison and ablation results are reported as mean and standard deviation over five random seeds under the same sampling protocol. For comparisons with published baselines, paired per-seed outputs are not available for all methods, so we report descriptive statistics without additional pairwise tests.

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TABLE V
CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE HOUSTON DATASET

Class	CACL	OSC-SCL	SSMLP-RPL	SSMTL	HyperCASR	MDL4OW	DBDA	SSFTT	SS-MTr	DSFormer	DSCP-Net
1	79.09±7.72	87.75±6.89	81.45±4.89	85.13±5.94	76.07±7.95	**93.33±5.13**	90.70±5.98	80.13±2.12	0.00±0.00	93.03±5.13	92.49±6.04
2	71.62±24.32	81.57±9.91	49.77±34.23	94.65±4.66	43.34±31.18	96.24±0.94	90.06±5.68	89.84±11.26	0.00±0.00	**96.69±4.90**	89.82±14.12
3	8.18±12.15	96.51±2.23	74.80±41.96	98.43±1.75	86.35±30.35	**99.97±0.06**	96.58±3.11	99.85±0.33	0.00±0.00	99.11±0.85	98.61±1.58
4	59.22±36.86	91.14±4.77	71.41±24.34	88.87±7.13	77.99±10.41	95.92±2.67	88.87±2.36	84.17±6.65	0.00±0.00	**97.71±2.19**	93.20±4.77
5	53.85±23.56	73.50±40.48	84.68±7.52	99.85±0.33	**100.00±0.00**	98.94±1.11	96.96±2.81	99.85±0.18	46.02±44.72	99.58±0.34	98.64±0.81
6	87.28±1.79	60.39±31.87	70.75±11.98	41.64±27.86	74.75±3.28	97.48±2.24	87.49±1.95	96.98±5.50	0.00±0.00	**98.33±1.81**	87.93±7.05
7	18.70±30.97	54.42±36.02	49.46±19.71	70.42±23.63	85.64±9.75	89.04±4.58	70.22±4.11	67.26±3.93	0.00±0.00	**93.05±1.17**	65.61±12.37
8	57.09±20.30	26.23±15.99	12.89±9.92	40.75±13.01	45.82±12.78	65.77±4.00	78.29±4.42	77.34±10.46	0.00±0.00	**87.59±4.97**	55.74±19.69
9	10.44±10.20	77.58±7.57	65.13±15.80	56.09±28.20	67.50±12.67	86.41±2.60	80.02±4.97	42.94±13.91	0.03±0.07	**87.48±2.52**	76.49±9.24
10	10.44±18.76	24.62±24.22	16.30±15.51	59.69±25.84	31.38±20.03	90.89±4.05	70.29±13.31	59.02±15.41	20.00±44.72	**94.59±3.48**	79.02±29.90
11	38.11±38.47	42.49±33.78	54.04±13.16	87.72±8.36	58.93±12.68	84.71±9.91	70.03±4.14	53.22±12.06	2.98±6.66	**95.97±1.13**	84.67±2.50
12	24.12±24.26	22.82±24.36	13.17±19.71	77.54±11.23	21.30±17.98	89.07±2.05	84.73±5.85	81.24±10.49	65.74±42.19	**89.76±5.04**	77.46±16.56
UnknownAcc	39.23±15.10	12.40±22.78	11.47±14.66	62.59±10.86	1.44±1.20	14.22±5.47	6.58±1.43	4.56±4.38	0.23±0.52	3.95±1.68	**85.28±9.53**
OA	41.35±6.12	55.29±6.34	47.45±2.74	74.97±5.60	56.10±6.12	81.97±0.30	74.87±0.62	67.94±2.11	11.10±2.93	**84.47±0.84**	82.34±8.01
AA	42.87±6.09	61.59±5.75	53.65±2.51	75.07±4.46	64.09±7.20	84.77±0.35	77.76±0.56	77.65±2.29	11.23±2.90	**87.45±0.90**	83.31±7.44
Kappa	35.76±6.85	51.33±6.86	43.07±2.80	72.71±6.09	52.23±6.74	80.40±0.32	72.67±0.67	65.16±2.27	3.06±3.17	**83.13±0.92**	80.76±8.72
HOS	37.92±8.59	14.69±23.60	15.64±15.30	68.02±8.45	2.76±2.23	23.76±8.35	12.16±2.47	8.17±7.42	0.43±0.96	7.53±3.11	**83.76±8.60**

In DSCP-Net, the router maps routing-space features to expert assignment probabilities. Unless otherwise specified, the main experiments use $M = 6$ experts and Top-$K = 2$ sparse routing for all datasets. Gaussian noise with $\sigma = 0.3$ is added to the router logits during training, and the Softmax temperature is set to $\tau_r = 1.0$. Each known class maintains three running prototypes in each feature space ($K_p = 3$), and the prototypes are updated using an exponential moving average with momentum $\mu = 0.99$. The differentiable auxiliary-objective weights in (13) are $\lambda_{\text{proto}} = 0.1$, $\lambda_{\text{align}} = 0.3$, and $\lambda_{\text{bal}} = 0.05$. The model is optimized with AdamW using an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}. We use a cosine annealing warm-restart scheduler with $T_0 = 50$ and $T_{\text{mult}} = 2$. The model is trained for 200 epochs with a batch size of 64.

During open-set inference, the unknown score is the average prototype distance across the two feature spaces. The score is normalized using z-score statistics estimated from the known-class validation set, and the global rejection threshold is set to the 95th percentile of the normalized validation scores by default. This default setting is used for all datasets in the main comparison to avoid dataset-specific tuning on the test set. A test sample is rejected as unknown if its normalized score exceeds the threshold; otherwise, it is assigned to the known class with the largest classification logit.

C. Compared Methods and Evaluation Metrics

We compare DSCP-Net with representative closed-set HSI classifiers, general OSR methods, and recent HSI open-set methods. The compared methods include CACL, OSC-SCL, SSMLP-RPL, SSMTL, HyperCASR, MDL4OW, DBDA, SSFTT, SS-MTr, and DSFormer [14], [32], [33], [50], [60]. To avoid using unknown test samples for threshold tuning, all compared methods follow the same split files, labeled training-calibration pools, and test sets as DSCP-Net. Methods that provide an open-set score use their native score when available. For methods originally designed for closed-set classification, we use the MSP as the rejection score to construct open-set baselines. In these MSP-based baselines, validation samples from known classes are used to estimate the confidence distribution, and a sample is rejected as unknown when its MSP falls below the validation-based threshold. Otherwise, it is assigned to the known class with the highest probability. Therefore, DSCP-Net and all baselines use only known-class validation information for threshold selection, without accessing any unknown test sample.

All methods are evaluated under the same training samples and testing splits. The evaluation metrics include open-set overall accuracy (OA), average accuracy (AA), Kappa coefficient (Kappa), unknown-class accuracy (UnknownAcc), and the harmonic mean of open-set performance (HOS). OA is computed over both known and unknown test samples under the open-set decision rule, AA is averaged over the known classes, Kappa is computed after assigning rejected samples to the unknown category, and UnknownAcc measures the proportion of unknown samples correctly rejected. HOS is used as the main metric because it reflects both overall OSR and unknown-class rejection

$$\text{HOS} = \frac{2 \times \text{OA} \times \text{UnknownAcc}}{\text{OA} + \text{UnknownAcc}}. \tag{22}$$

For multiseed experiments, each metric, including HOS, is computed for each seed and then reported as mean ± standard deviation across seeds.

D. Quantitative Comparison

Tables V–VIII report the complete per-class accuracies and aggregate metrics on the four datasets. DSCP-Net obtains HOS values of 83.76%, 56.15%, 93.71%, and 93.05% on Houston, IP, LK, and UP, respectively, showing a consistent balance between retaining known samples and rejecting unknown ones. The qualitative classification maps of the compared methods on the Houston, Indian Pines, WHU-Hi-LongKou, and Pavia University datasets are presented in Figs. 8–11, respectively.

On the Houston dataset, DSCP-Net achieves an HOS of 83.76%, with the highest UnknownAcc of 85.28% among the recalculated open-set results and a high OA of 82.34%. Among the recalculated baselines, SSMTL provides the strongest open-set balance with an HOS of 68.02%, but its UnknownAcc remains lower than that of DSCP-Net.

On IP, which contains highly similar agricultural categories and severe spectral confusion, DSCP-Net obtains an OA of

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TABLE VI

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE IP DATASET

Class	CACL	OSC-SCL	SSMLP-RPL	SSMTL	HyperCASR	MDL4OW	DBDA	SSFTT	SS-MTr	DSFormer	DSCP-Net
1	**100.00±0.00**	49.23±47.37	53.08±24.08	95.38±10.32	34.62±27.47	99.57±0.87	**100.00±0.00**	95.38±8.34	73.08±43.51	**100.00±0.00**	99.23±1.72
2	63.81±23.55	6.35±10.95	50.28±23.53	76.95±6.20	42.39±37.99	86.85±5.75	72.61±7.16	40.06±18.55	13.86±31.00	**91.02±2.80**	73.93±5.84
3	54.20±32.34	91.53±9.84	46.47±31.22	89.93±11.10	32.12±17.02	84.99±1.29	78.44±4.22	46.02±17.60	32.94±46.80	**92.32±4.37**	83.75±9.64
4	90.88±11.93	28.02±40.31	44.15±13.93	77.88±12.73	36.41±41.47	91.14±3.43	88.82±4.44	**100.00±0.00**	0.00±0.00	97.85±2.62	94.47±5.79
5	68.55±35.44	29.24±38.39	61.56±3.90	92.48±5.19	50.67±13.75	83.98±3.97	**93.00±4.02**	76.50±8.94	0.00±0.00	91.80±3.13	82.94±10.25
6	93.69±5.06	70.85±39.99	87.18±5.71	**100.00±0.00**	73.24±22.94	98.77±0.72	99.59±0.37	98.17±1.10	1.24±2.77	99.25±0.74	93.38±3.18
7	55.00±45.14	40.00±54.77	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	0.00±0.00	**100.00±0.00**	**100.00±0.00**
8	70.00±30.46	60.57±39.05	37.47±26.61	69.65±10.65	66.07±21.08	99.83±0.33	98.64±1.03	95.33±10.45	18.17±40.62	**100.00±0.00**	93.71±3.29
UnknownAcc	26.89±11.49	11.34±12.95	4.66±1.46	2.43±1.59	3.77±4.36	8.43±2.69	1.15±0.35	**48.59±12.36**	12.06±8.30	1.62±0.21	44.33±16.74
OA	44.39±4.29	24.81±11.04	25.34±2.09	35.85±1.39	22.16±7.45	42.38±1.58	35.11±1.39	55.35±9.95	12.84±3.79	39.25±0.64	**83.77±2.38**
AA	69.22±14.42	46.97±27.05	60.02±3.53	87.78±2.42	54.44±12.98	83.73±0.77	81.36±1.68	81.43±4.00	17.41±7.02	85.98±0.80	**90.18±2.09**
Kappa	30.59±4.73	14.54±9.62	49.57±5.85	28.62±0.88	13.87±6.79	34.33±1.26	28.68±1.52	40.65±8.98	3.21±0.81	32.74±0.79	**80.24±2.65**
HOS	32.34±9.64	13.93±13.95	7.77±2.08	4.48±2.75	5.69±6.36	13.90±3.77	2.27±0.67	51.65±11.30	11.03±6.48	3.19±0.41	**56.15±15.87**

TABLE VII

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE LK DATASET

Class	CACL	OSC-SCL	SSMLP-RPL	SSMTL	HyperCASR	MDL4OW	DBDA	SSFTT	SS-MTr	DSFormer	DSCP-Net
1	95.40±3.66	91.97±6.73	93.98±2.83	92.71±4.93	98.56±1.87	99.21±0.32	99.34±0.98	83.41±7.46	3.69±8.22	**99.95±0.05**	96.09±2.16
2	93.49±3.64	60.84±33.80	8.87±8.84	77.93±6.76	53.91±46.09	**98.17±0.87**	79.64±19.76	97.10±1.28	29.87±42.98	97.40±1.17	89.51±10.56
3	97.07±1.31	58.13±39.91	56.29±39.43	99.01±1.22	91.52±12.89	98.24±0.66	91.51±14.70	**99.82±0.22**	37.05±51.00	99.69±0.43	93.12±3.62
4	71.15±13.23	34.00±31.35	75.50±7.96	77.17±24.09	67.04±27.33	91.33±3.16	83.41±3.80	75.89±10.41	49.84±46.87	**93.14±2.05**	85.06±3.90
5	93.03±13.04	38.03±27.65	81.43±15.17	92.80±8.31	97.07±3.51	93.61±3.31	63.34±31.83	47.85±25.53	52.06±48.13	**98.39±0.58**	91.49±6.96
6	90.14±5.62	80.16±21.13	88.58±2.24	84.55±11.14	98.24±1.31	96.46±0.79	95.09±4.43	92.00±6.35	27.48±34.49	**99.56±0.32**	94.31±3.70
7	97.23±2.27	78.75±44.03	**99.97±0.04**	99.09±0.76	59.98±26.51	97.41±0.36	99.83±0.24	48.65±20.97	98.56±2.05	99.85±0.11	99.43±0.58
8	72.87±20.97	83.23±13.33	46.04±29.96	57.09±10.80	85.22±11.31	84.96±5.43	82.33±6.89	**99.93±0.07**	6.61±14.68	95.53±2.93	81.17±4.84
UnknownAcc	30.40±10.44	39.18±14.09	45.86±3.33	12.14±7.60	0.00±0.01	64.39±7.40	10.57±2.85	1.33±1.42	10.71±4.50	2.41±0.91	**94.92±5.02**
OA	85.65±5.82	64.51±14.05	82.73±1.02	85.72±7.87	71.20±10.78	94.47±0.94	89.82±1.55	68.74±8.89	53.27±11.59	**95.00±0.59**	92.66±1.35
AA	82.31±4.79	65.64±8.03	68.83±5.67	85.04±4.82	81.44±7.14	**91.53±0.63**	78.34±4.01	80.58±5.74	38.15±7.59	87.32±0.28	91.27±2.04
Kappa	81.91±7.04	58.11±14.86	77.86±1.05	81.11±10.79	65.47±11.37	92.82±1.19	86.83±1.93	62.28±9.96	41.02±12.18	**93.47±0.75**	90.42±1.70
HOS	43.86±10.59	47.85±14.42	58.94±2.64	20.61±11.86	0.01±0.02	76.35±4.92	18.85±4.57	2.58±2.73	17.14±5.59	4.69±1.73	**93.71±2.44**

TABLE VIII

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE UP DATASET

Class	CACL	OSC-SCL	SSMLP-RPL	SSMTL	HyperCASR	MDL4OW	DBDA	SSFTT	SS-MTr	DSFormer	DSCP-Net
1	65.69±32.56	71.94±18.16	79.11±9.54	51.64±35.21	62.01±10.06	75.30±10.63	67.59±4.35	66.06±30.27	0.00±0.00	**94.88±3.23**	82.39±9.21
2	64.84±17.93	56.42±38.74	48.49±18.20	66.46±26.19	35.95±8.21	68.57±14.41	90.63±8.03	74.77±9.02	6.28±14.05	**95.12±4.03**	90.22±4.55
3	95.95±2.31	46.65±29.69	33.04±27.05	68.71±34.78	77.41±11.13	50.05±21.59	75.28±13.36	90.44±8.42	0.00±0.00	**96.24±1.10**	81.39±9.63
4	95.68±2.40	67.80±38.34	70.53±22.90	78.32±19.47	91.23±5.42	79.52±10.92	96.28±1.01	87.89±5.56	9.40±21.01	**96.62±2.02**	84.73±8.68
5	99.89±0.13	95.02±4.89	93.46±3.26	23.25±15.96	**100.00±0.00**	68.33±17.64	99.77±0.25	**100.00±0.00**	0.00±0.00	**100.00±0.00**	97.01±4.90
6	78.67±12.53	27.48±29.10	52.73±18.10	85.15±21.54	70.45±22.85	54.75±19.26	75.78±13.70	77.50±14.52	79.03±44.23	**97.95±1.47**	90.05±4.45
7	79.11±16.67	86.92±10.22	34.70±29.26	99.56±0.61	92.95±9.56	63.24±11.27	82.32±13.45	98.38±1.24	0.00±0.00	**99.75±0.16**	94.87±2.82
8	42.63±28.09	82.58±7.20	52.38±25.46	45.91±38.44	72.22±31.03	29.60±30.80	60.51±10.93	43.64±28.42	20.00±44.72	**94.20±3.33**	84.27±5.11
UnknownAcc	41.35±21.91	16.28±30.28	3.27±5.97	0.00±0.00	0.00±0.00	0.23±0.28	0.05±0.09	7.07±7.11	6.21±6.03	10.52±8.33	**98.90±0.90**
OA	69.42±7.98	59.25±21.14	54.86±6.33	63.74±12.85	56.08±6.40	62.83±4.96	80.39±2.65	72.78±6.36	14.56±6.49	**93.92±1.69**	87.95±4.34
AA	73.76±2.81	66.85±9.66	58.05±2.08	64.87±7.29	75.28±5.59	61.17±3.78	72.02±1.47	79.84±4.83	14.34±2.56	87.25±0.96	**88.12±4.31**
Kappa	62.75±8.51	51.03±20.47	44.65±5.27	55.38±13.51	49.21±6.61	52.80±4.81	74.26±3.12	66.01±7.50	3.40±4.41	**92.00±2.15**	84.27±5.46
HOS	47.31±15.43	18.04±29.84	5.28±9.15	0.00±0.00	0.00±0.00	0.46±0.56	0.09±0.19	11.99±10.52	6.79±6.35	17.91±13.12	**93.05±2.16**

83.77%, an UnknownAcc of 44.33%, and an HOS of 56.15%. SSFTT obtains a comparable UnknownAcc of 48.59%, but its OA and Kappa are lower, leading to a lower HOS of 51.65%. This result shows that IP remains the most difficult split for unknown rejection because some unknown agricultural categories lie close to the known-class manifold.

On LK, DSCP-Net reaches an HOS of 93.71%, with an OA of 92.66% and an UnknownAcc of 94.92%. The strongest recalculated baseline is SSMLP-RPL, with an HOS of 58.94%. The high UnknownAcc and OA indicate that calibrated prototype distances separate unknown samples well without rejecting many known samples.

On UP, DSCP-Net achieves a strong open-set balance, with an OA of 87.95%, an UnknownAcc of 98.90%, and an HOS of 93.05%. Among the recalculated baselines, OSC-SCL provides the highest HOS of 18.04%, while several baselines preserve known-class accuracy but reject very few unknown samples. This result indicates that known-class accuracy alone is not sufficient for evaluating open-set hyperspectral recognition.

DSFormer makes this tradeoff clear. Its OA is higher than that of DSCP-Net on Houston, LK, and UP (84.47%, 95.00%, and 93.92%, respectively), and its AA or Kappa is also strong because the known-class classifier retains effective within-set decision boundaries. Yet its UnknownAcc is only 3.95%, 2.41%, and 10.52% on these datasets, which means that MSP-based rejection still assigns most unknown samples to known classes. Its HOS is correspondingly low at 7.53%, 4.69%, and 17.91%. DSCP-Net sacrifices some accuracy on metrics dominated by known-class recognition but rejects unknown samples much more reliably. Higher OA, AA, or Kappa for DSFormer is therefore compatible with a much lower HOS. The two methods favor different operating points, and we do not claim that DSCP-Net is better on every closed-set-oriented metric.

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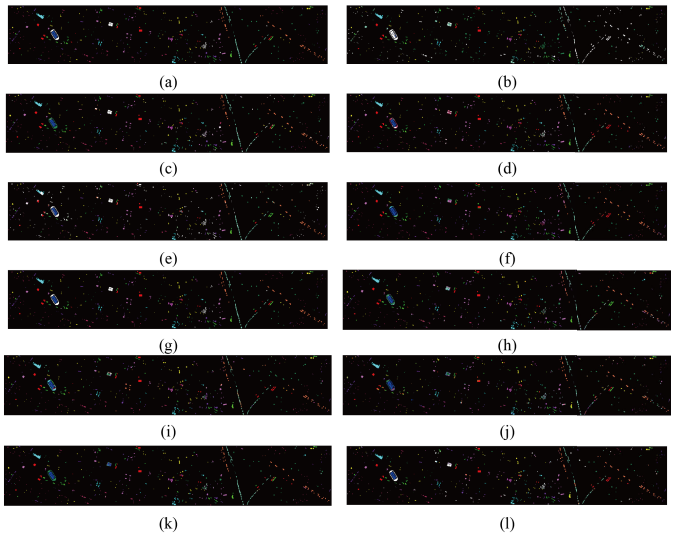


Fig. 8. Classification maps of different methods on the Houston dataset. (a) Ground-truth map. (b) CACL. (c) OSC-SCL. (d) SSMLP-RPL. (e) SSMTL. (f) HyperCASR. (g) MDL4OW. (h) DBDA. (i) SSFTT. (j) SS-MTr. (k) DSFormer. (l) DSCP-Net.

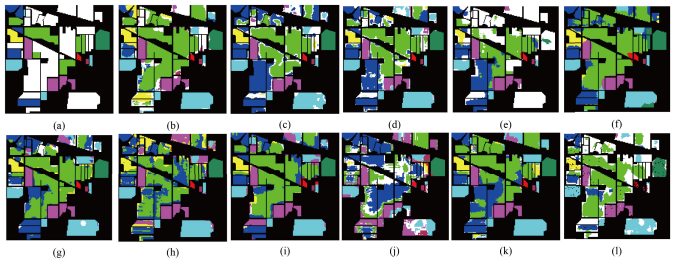


Fig. 9. Classification maps of different methods on the IP dataset. (a) Ground-truth map. (b) CACL. (c) OSC-SCL. (d) SSMLP-RPL. (e) SSMTL. (f) HyperCASR. (g) MDL4OW. (h) DBDA. (i) SSFTT. (j) SS-MTr. (k) DSFormer. (l) DSCP-Net.

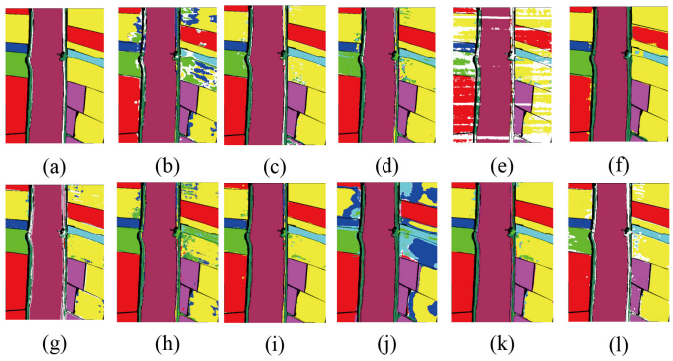


Fig. 10. Classification maps of different methods on the LK dataset. (a) Ground-truth map. (b) CACL. (c) OSC-SCL. (d) SSMLP-RPL. (e) SSMTL. (f) HyperCASR. (g) MDL4OW. (h) DBDA. (i) SSFTT. (j) SS-MTr. (k) DSFormer. (l) DSCP-Net.

IP exposes a limitation of the current framework. Although DSCP-Net achieves the best HOS on this dataset, its UnknownAcc remains lower than those on Houston, LK, and UP. Distance-based rejection is harder when unknown samples lie close to known agricultural categories in the learned feature space. DSCP-Net therefore improves the open-set balance, but it does not fully solve unknown rejection under severe spectral confusion. This limitation motivates future work on class-wise calibration, local density estimation, and more robust representation learning that explicitly accounts for unknown classes.

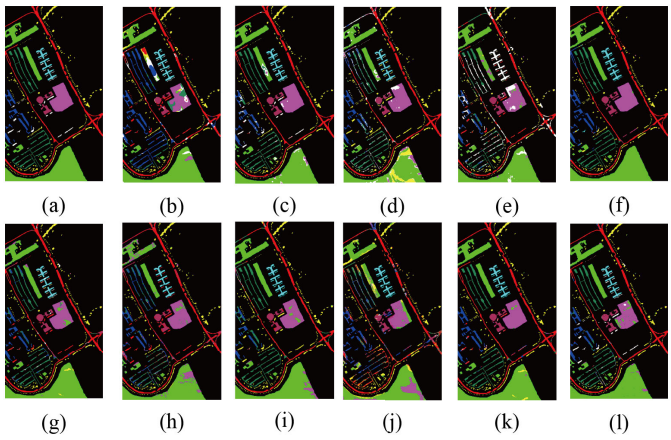


Fig. 11. Classification maps of different methods on the UP dataset. (a) Ground-truth map. (b) CACL. (c) OSC-SCL. (d) SSMLP-RPL. (e) SSMTL. (f) HyperCASR. (g) MDL4OW. (h) DBDA. (i) SSFTT. (j) SS-MTr. (k) DSFormer. (l) DSCP-Net.

E. Visualization Analysis

Fig. 12 shows the distributions of calibrated unknown scores for known and unknown test samples. Known samples are mainly distributed in the low-score region, whereas unknown samples tend to shift toward larger scores, consistent with the open-set decision rule of DSCP-Net. On Houston, LK, and UP, the score distributions of known and unknown samples are relatively well separated, and the validation-based threshold lies near the transition area between them. This explains the high UnknownAcc and HOS achieved by DSCP-Net on these datasets. By contrast, the score distributions on IP show stronger overlap, indicating that the boundary between difficult known samples and unknown samples is less distinct. This is consistent with the severe spectral confusion among agricultural categories in IP and explains the lower unknown rejection accuracy and larger result variance on this dataset.

The t-SNE visualization in Fig. 13 gives a qualitative view of the learned feature distributions. On relatively separable datasets such as LK and UP, known-class samples form compact clusters, and unknown samples tend to appear outside the cluster cores or form separate groups. This pattern agrees with the distance-based rejection mechanism of DSCP-Net. On Houston and IP, the distributions are more scattered because of spectral confusion and complex land-cover composition. The overlap on IP helps explain its less stable unknown rejection. As t-SNE is qualitative, the visualization is used only as supporting evidence for the quantitative results in Tables V–VIII.

F. Parameter Sensitivity Analysis

The main comparison uses the same default setting, $q = 0.95$, across all datasets. This section analyzes three factors

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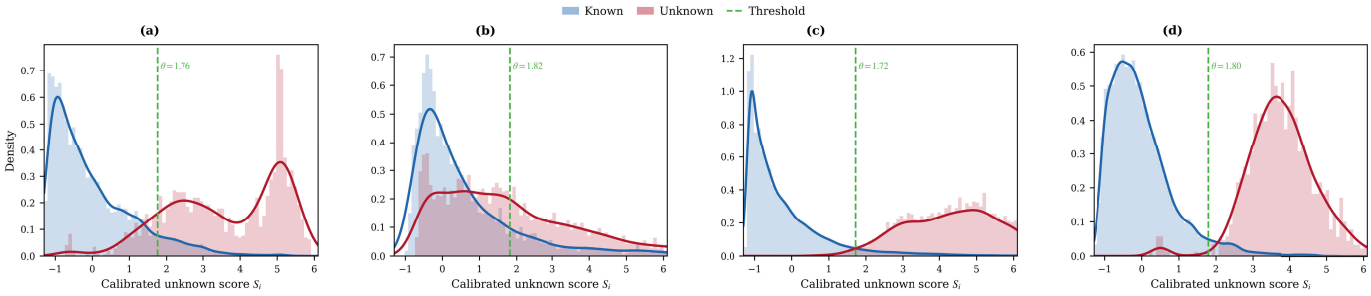


Fig. 12. Distributions of calibrated unknown scores for known and unknown test samples on four hyperspectral datasets. The dashed vertical line denotes the validation-based rejection threshold. Better separation between the two distributions indicates a more reliable open-set decision boundary. (a) Houston. (b) IP. (c) LK. (d) UP.

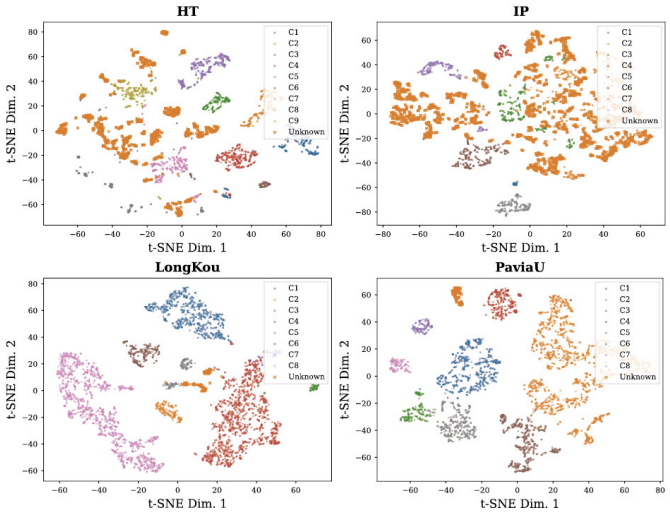


Fig. 13. t-SNE visualization of learned features on the four hyperspectral datasets.

that may affect the open-set balance: the MoE structure, including the number of experts M and activated experts K; the threshold quantile q used in validation-based global calibration; and the number of prototypes K_p for each known class. HOS is used as the main sensitivity metric.

1) Effect of the Number of Experts and Top-K Routing: The MoE module is controlled by two structural hyperparameters: the number of experts M and the number of activated experts K for each input sample. We perform a grid search with $M \in \{2, 4, 6, 8, 10\}$ and $K \in \{1, 2, 3, 4, 5\}$, excluding invalid settings with $K > M$. The results are shown in Fig. 14.

The highest observed HOS depends on the dataset: Houston favors sparse routing with a larger expert pool ($M = 8$, $K = 1$), while IP and LK benefit from denser activation ($M = 6$, $K = 4$ or 5). UP reaches the highest observed HOS with $M = 8$ and $K = 3$. These results indicate that the MoE structure is a tradeoff between representation flexibility and routing selectivity, rather than a component whose benefit grows monotonically with more experts or denser activation.

2) Effect of the Threshold Quantile: The threshold quantile q controls the rejection boundary in the global calibration stage. DSCP-Net uses the qth quantile of the validation score distribution as the global rejection threshold. A larger q usually produces a more conservative rejection boundary, preserving more known-class samples but reducing sensitivity to unknown samples. We vary q in $\{0.93, 0.95, 0.97, 0.99\}$ and report the results in Fig. 15.

The effect of q differs across scenes. On Houston and IP, lower quantiles produce better HOS. The best HOS is obtained at $q = 0.93$ on both datasets, and increasing q gradually reduces UnknownAcc. Datasets with stronger spectral confusion and more complex unknown distributions therefore need a stricter rejection boundary to avoid absorbing unknown samples into known categories. By contrast, LK and UP benefit from higher quantiles. Their HOS values increase as q becomes larger, while UnknownAcc remains relatively high. For these more separable scenes, a conservative threshold can preserve known-class samples without greatly reducing unknown rejection.

This behavior motivates validation-based calibration. Because prototype-distance distributions vary with sensor characteristics, spatial resolution, and class composition, the quantile parameter q provides an interpretable control factor: a smaller q produces a stricter rejection boundary, whereas a larger q preserves more difficult known samples. In the main experiments, $q = 0.95$ is used as a unified default rather than a dataset-specific optimum.

3) Effect of the Number of Prototypes: The number of prototypes K_p controls the granularity of known-class structure modeling. Fig. 16 shows that multiple prototypes are usually helpful, especially on Houston and IP, where intraclass variation and confusion between known and unknown classes are stronger. On Houston, HOS increases from 68.22% at $K_p = 1$ to 86.53% at $K_p = 5$; on IP, it increases from 56.45% to 65.93%. The changes are milder on LK and UP. We therefore use $K_p = 3$ as a balanced default rather than a dataset-specific optimum.

In deployment, these hyperparameters can be selected from the known-class training–validation pool without labeled unknown samples. Given an acceptable known-class false-rejection budget α, we set $q = 1 - \alpha$ because approximately $1 - q$ of the known validation scores lie above the qth quantile before finite-sample effects. Candidate values of M, K, and K_p are then compared using known-class validation accuracy and stability across repeated splits, subject to memory and latency constraints. We apply a one-standard-error rule and choose the

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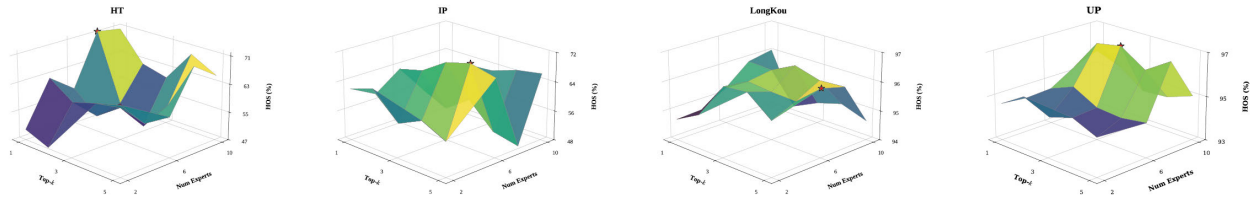


Fig. 14. Sensitivity analysis of the number of experts M and Top-K routing in terms of HOS. The red star denotes the highest observed HOS in this diagnostic sweep for each dataset; it is not a test-time hyperparameter-selection rule. Gray cells indicate invalid settings where $K > M$.

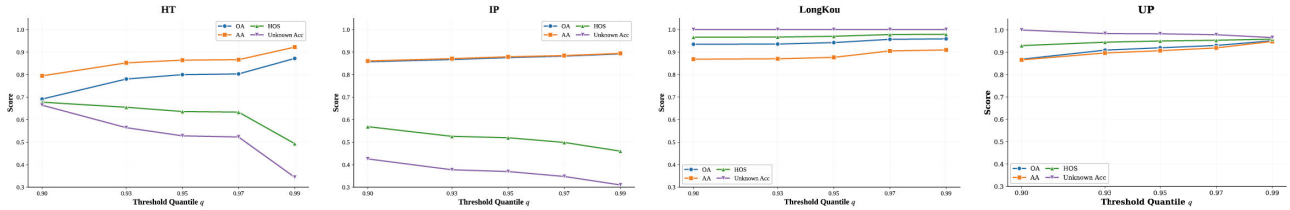


Fig. 15. Sensitivity analysis of the threshold quantile q used in validation-based global calibration. The left reports HOS, and the right reports UnknownAcc.

TABLE IX

COMPUTATIONAL COMPLEXITY COMPARISON WITH ALL COMPARED METHODS ON THE FOUR DATASETS. PARAMS DENOTES LEARNABLE PARAMETERS IN MILLIONS. GFLOPS IS MEASURED FOR ONE FORWARD PASS UNDER THE CORRESPONDING DATASET CONFIGURATION

Dataset	Metric	CACL	OSC-SCL	SSMLP-RPL	SSMTL	HyperCASR	MDL4OW	DBDA	SSFTT	SS-MTr	DSFormer	DSCP-Net
Houston	Params	0.108	0.538	5.526	2.236	0.684	0.356	0.248	0.153	0.147	0.677	0.849
	GFLOPs	0.0363	0.3044	6.9612	0.3610	0.4928	0.0474	0.0388	0.0228	0.0097	0.0407	0.0642
Indian Pines	Params	0.112	0.736	10.459	2.856	0.798	0.420	0.282	0.153	0.151	0.676	0.850
	GFLOPs	0.0376	0.4267	11.9162	0.4615	0.5742	0.0558	0.0443	0.0228	0.0105	0.0407	0.0645
LongKou	Params	0.116	0.983	18.884	3.631	0.819	0.501	0.324	0.153	0.155	0.676	0.852
	GFLOPs	0.0391	0.5797	19.8802	0.5870	0.5892	0.0663	0.0512	0.0228	0.0115	0.0407	0.0648
PaviaU	Params	0.105	0.397	2.932	1.782	0.636	0.308	0.223	0.153	0.144	0.676	0.847
	GFLOPs	0.0355	0.2170	4.1316	0.2875	0.4584	0.0413	0.0347	0.0228	0.0091	0.0407	0.0640

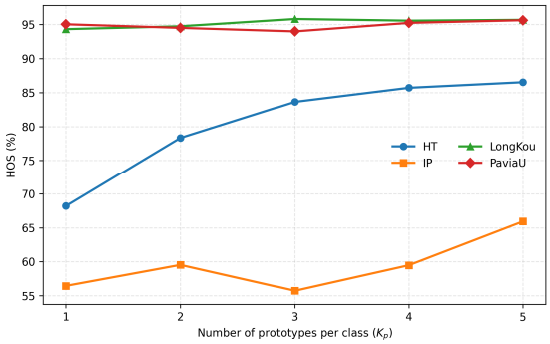


Fig. 16. Sensitivity analysis of the number of prototypes per known class K_p in terms of HOS.

smaller configuration when its mean validation performance is within one standard error of the best candidate. This procedure does not use unknown test labels, but without representative labeled unknown validation samples it cannot guarantee the HOS-optimal configuration for a new deployment distribution. The sensitivity plots are diagnostic and are not used to tune the model on unknown test labels. If application-specific validation is too limited for a reliable search, the main settings $M = 6$, $K = 2$, $K_p = 3$, and $q = 0.95$ provide a reproducible starting point; the threshold should be recalibrated if the acquisition conditions change.

4) Computational Complexity: Table IX reports the computational cost of DSCP-Net and all compared methods on the four datasets. GFLOPs is used instead of raw wall-clock time because the compared methods rely on heterogeneous implementations and hardware settings. DSCP-Net has about 0.85M learnable parameters across datasets, and its forward cost remains below 0.07 GFLOPs per patch. The learnable parameter count mainly comes from dual-space projections and sparse expert transformations; the prototype buffers are nonlearnable state. Under the same dataset-specific profiling protocol, the GFLOPs of DSCP-Net are much lower than those of SSMLP-RPL, SSMTL, and HyperCASR, suggesting a moderate computational cost for few-shot open-set HSI classification.

G. Ablation Study

We use two ablation tables to separate changes to the system from changes to the training objective. Table X removes the

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TABLE X

SYSTEM-COMPONENT ABLATION OF DSCP-NET IN TERMS OF HOS. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE SEEDS. THE VARIANTS REMOVE THE MOE-BASED CONDITIONAL TRANSFORMATION, MULTIPROTOTYPE MODELING, OR VALIDATION-BASED CALIBRATION WHILE PRESERVING THE REMAINING SYSTEM

Dataset	Full	No MoE	No MP	No Calib.
HT	83.76 ± 8.60	83.40 ± 9.21	78.96 ± 9.93	78.71 ± 7.93
IP	56.15 ± 15.87	56.42 ± 13.10	51.79 ± 14.14	45.14 ± 10.04
LK	93.71 ± 2.44	92.45 ± 4.42	90.66 ± 4.17	77.38 ± 7.46
UP	93.05 ± 2.16	91.74 ± 4.36	93.10 ± 3.11	90.17 ± 9.09

TABLE XI

OBJECTIVE-LEVEL ABLATION CORRESPONDING TO THE GRADIENT-BASED OBJECTIVE IN (13), REPORTED IN TERMS OF HOS AS MEAN ± STANDARD DEVIATION OVER FIVE SEEDS. EACH "W/O" ROW SETS ONLY THE SPECIFIED DIFFERENTIABLE AUXILIARY-OBJECTIVE WEIGHT TO ZERO WHILE PRESERVING ALL ARCHITECTURAL AND INFERENCE COMPONENTS

Objective variant	HT	IP	LK	UP
Full	83.76 ± 8.60	56.15 ± 15.87	93.71 ± 2.44	93.05 ± 2.16
w/o $\mathcal{L}_{\text{proto}}$	85.86 ± 5.95	53.83 ± 13.32	92.44 ± 4.52	92.91 ± 2.17
w/o $\mathcal{L}_{\text{align}}$	84.88 ± 7.32	55.88 ± 16.00	93.65 ± 2.37	93.64 ± 1.79
w/o $\mathcal{L}_{\text{bal}}$	84.12 ± 8.62	56.19 ± 16.07	93.74 ± 2.37	93.21 ± 1.96
$\mathcal{L}_{\text{cls}}$ only	85.89 ± 6.05	53.24 ± 12.21	92.46 ± 4.46	93.16 ± 1.95

MoE-based conditional transformation, multiprototype modeling, or validation-based calibration while leaving the rest of the system unchanged. The dual-space representation is the base inference design of DSCP-Net, so we discuss it through the method motivation and inference analysis rather than introducing routing-only or image-only replacements. Table XI instead tests the differentiable terms in (13) by setting one auxiliary weight to zero at a time and by including an $\mathcal{L}_{\text{cls}}$-only reference.

The two tables therefore answer different questions. "No MP" in Table X removes both the multiprototype representation and its inference mechanism, whereas "w/o $\mathcal{L}_{\text{proto}}$" in Table XI retains the representation and removes only its training loss. The $\mathcal{L}_{\text{cls}}$-only variant also retains the complete dual-space encoder, MoE transformation, EMA multiprototype representation, and validation-calibrated prototype-distance inference; only the differentiable auxiliary terms are disabled. Both tables report HOS because OSR depends on known-class classification and unknown rejection. We interpret the results as dataset-specific marginal effects rather than evidence that the terms are uniformly additive.

Table X shows that the effect of each system component depends on the dataset. Removing MoE does not consistently lower HOS, which suggests that the conditional expert transformation is an auxiliary representation component rather than the main source of the open-set gain.

The multiprototype and calibration variants agree with the sensitivity analyses in Figs. 15 and 16. Removing multiprototype modeling changes HOS from 83.76 to 78.96 on Houston, from 56.15 to 51.79 on IP, and from 93.71 to 90.66 on LK, while UP remains essentially unchanged (93.05 to 93.10). Calibration has the largest and most consistent effect: without it, HOS changes from 83.76 to 78.71 on Houston, from 56.15 to 45.14 on IP, from 93.71 to 77.38 on LK, and from 93.05 to 90.17 on UP. These results support interpreting DSCP-Net primarily as a calibrated prototype-distance framework, with the effects of the other components varying by scene.

Table XI isolates the terms in (13). Compared with the $\mathcal{L}_{\text{cls}}$-only reference, the full objective raises HOS by 2.91 points on IP and 1.25 points on LK. The $\mathcal{L}_{\text{cls}}$-only variant is higher by 2.13 points on Houston and 0.11 points on UP. Removing $\mathcal{L}_{\text{proto}}$ lowers HOS by 2.32 and 1.27 points on IP and LK, respectively, but increases it on Houston. The changes produced by $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{bal}}$ are smaller and vary by dataset relative to the reported standard deviations. Table XI therefore describes the marginal effect of each training term, while Table X evaluates architecture and inference components and shows that validation-based calibration has the strongest and most consistent component-level effect.

V. CONCLUSION

This article proposed DSCP-Net for open-set HSI classification. DSCP-Net constructs structured known-class representations through dual-space feature learning, conditional expert transformation, and multiprototype modeling, and then rejects unknown samples using calibrated dual-space prototype distances. Z-score normalization and validation-based quantile calibration provide an adaptive rejection boundary across hyperspectral scenes.

Experiments on four public hyperspectral datasets show that DSCP-Net better balances known-class recognition and unknown-class rejection than the compared methods. Multiprototype modeling and, in particular, validation-based calibration make the largest component-level contributions to this balance. Among the training terms, the prototype contrastive loss helps most on IP and LK, whereas dual-space alignment and soft-routing importance balancing have smaller effects that vary by scene. Future work will study cross-scene threshold transfer, more reliable rejection for highly confused agricultural scenes such as IP, and OSR with large-scale remote sensing foundation models.

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